

Omkar Yadav

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EDUCATION

Sardar Patel Institute of Technology

B.Tech in Electronics and Telecommunication (CGPA: 9/10)

Mumbai, IN

Nov. 2022 – Present

EXPERIENCE

Summer Intern

IIT Bombay (Department of Mechanical Engineering)

Mumbai, IN

May. 2025– Present

- Built a reliability and mission validation module for Indian Navy ships. Generated **reliability trees** and automated **K:N** configuration checks to ensure mission feasibility, identify faulty components, and estimate spare part budgets.
- Developed a pipeline to convert PDF and Excel documents into a **Knowledge Graphs** using SciPhi/triplex, LLaMA3.3, and Neo4j. Enabled semantic question answering over naval equipment manuals using **GraphRAG**.

Software Development Intern

MoneyGrow India Asset Private Ltd

Mumbai, IN

Dec. 2023 – Jan. 2024

- Built Python automation tool using Selenium to process **100+** client's daily financial statements, improving operational efficiency by **90%**. Created Tkinter based GUI for non technical staff.
- Developed a WhatsApp bot using Flask and Twilio WhatsApp Business API for client document requests; implemented SQLite database for client lookup and deployed on AWS EC2 with automated email delivery.

PROJECTS

Samarth.ai | FastAPI, PostgreSQL, ChromaDB, Redis, LLaMa | [GitHub](#)

- Led backend and contributed in RAG development in a team of 6 to develop a chatbot, achieving **<1s** response times for 10+ concurrent users with **95% accuracy** on GAIL India evaluation dataset.
- Implemented **contextual chunk headers** strategy to improve retrieval context and response relevance.
- Built OCR pipeline for policy document ingestion and automated keyword extraction.
- Architected authentication system with JWT, 2FA, and Redis for LLM responses caching.

Gozam | Go, Gin, PostgreSQL, Redis, FFmpeg | [GitHub](#)

- Developed an audio recognition webapp in Go with Gin using various **Digital Signal Processing** techniques.
- Engineered an audio fingerprinting algorithm that converts spectral peaks into compact **32-bit hashes** using **frequency pairs and time deltas**, enabling robust song identification with **100ms** timing tolerance.
- Implemented a hybrid storage system with **Redis** for real-time fingerprint lookups and **PostgreSQL** for managing song metadata, achieving low-latency querying and reliable persistence.
- Optimized performance by downsampling input audio and restricting analysis to the **20Hz–5kHz** range, **reducing computational overhead** while enhancing signal-to-noise ratio.

TECHNICAL SKILLS

Languages: Python, Go, SQL, Java, HTML/CSS

Frameworks: FastAPI, Gin, Django, Django restframework, Flask, Bootstrap

Databases: PostgreSQL, MySQL, SQLite, Redis, Neo4j

Tools & Platforms: Git, Selenium, MATLAB, GitHub

ACHIEVEMENTS

Smart India Hackathon 2024 - Secured **1st place** among 170+ teams in a national-level competition.

HERE India Hackathon 2025 - Secured **Top 10** position among 40+ teams.

Achieved **Top 10** placements in **5** intercollegiate hackathons.

LEADERSHIP AND POSITION OF RESPONSIBILITY

Technical Head

IEEE Computer Society S.P.I.T.

Aug. 2024 – May 2025

Technical Assistant

Sardar Patel Institute of Technology

Feb. 2024 – Mar. 2024